



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Tower of Hanoi Problem

TOWER OF HANOI PROBLEM

"The Tower of Hanoi is a classic problem demonstrating recursion and stack behavior. Write a C program to solve the Tower of Hanoi problem and display the sequence of moves required to transfer all disks.

Input

Number of Disks = 3

Output

Move Disk 1 from A to C
Move Disk 2 from A to B
Move Disk 1 from C to B
Move Disk 3 from A to C
Move Disk 1 from B to A
Move Disk 2 from B to C
Move Disk 1 from A to C"

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 void towerOfHanoi(int n, char from_rod, char to_rod, char aux_rod)
4     if (n == 1) {
5         printf("Move Disk 1 from %c to %c\n", from_rod, to_rod);
6         return;
7     }
8     towerOfHanoi(n - 1, from_rod, aux_rod, to_rod);
9     printf("Move Disk %d from %c to %c\n", n, from_rod, to_rod);
10    towerOfHanoi(n - 1, aux_rod, to_rod, from_rod);
11 }
12
13 int main() {
14     int n;
15     printf("Number of Disks = ");
16     if (scanf("%d", &n) != 1) {
17         return 1;
18     }
19
20     towerOfHanoi(n, 'A', 'C', 'B');
21
```

OUTPUT

```
Number of Disks = Move Disk 1 from A to C
Move Disk 2 from A to B
Move Disk 1 from C to B
Move Disk 3 from A to C
Move Disk 1 from B to A
Move Disk 2 from B to C
Move Disk 1 from A to C
```

INPUT

3

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